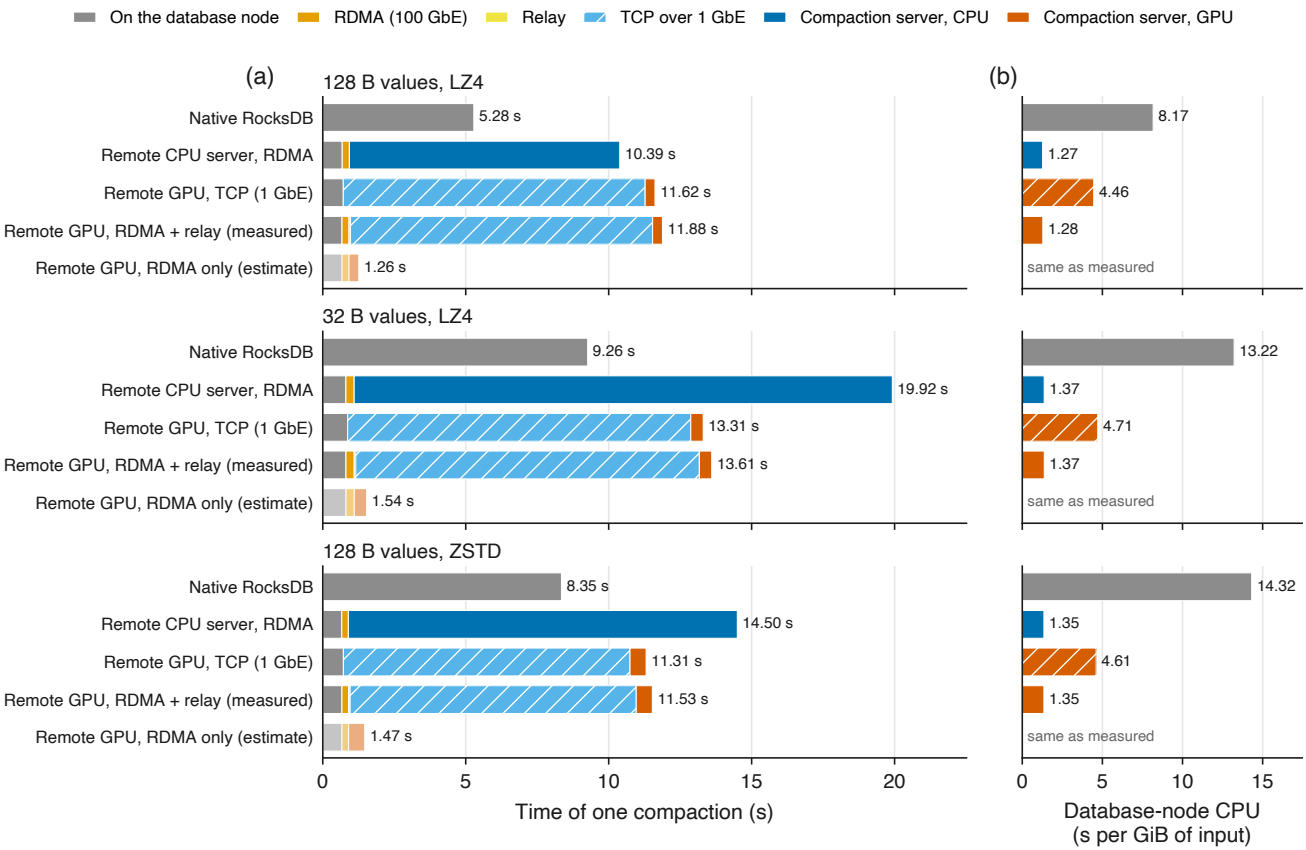
On the database node
RDMA (100 GbE)
Relay
TCP over 1 GbE
Compaction server, CPU
Compaction server, GPU
(a)
128 B values, LZ4
Native RocksDB
5.28 s
Remote CPU server, RDMA
10.39 s
Remote GPU, TCP (1 GbE)
11.62 s
Remote GPU, RDMA + relay (measured)
11.88 s
Remote GPU, RDMA only (estimate)
1.26 s
32 B values, LZ4
Native RocksDB
9.26 s
Remote CPU server, RDMA
19.92 s
Remote GPU, TCP (1 GbE)
13.31 s
Remote GPU, RDMA + relay (measured)
13.61 s
Remote GPU, RDMA only (estimate)
1.54 s
128 B values, ZSTD
Native RocksDB
8.35 s
Remote CPU server, RDMA
14.50 s
Remote GPU, TCP (1 GbE)
11.31 s
Remote GPU, RDMA + relay (measured)
11.53 s
Remote GPU, RDMA only (estimate)
1.47 s
0
5
10
15
20
Time of one compaction (s)
(b)
8.17
1.27
4.46
1.28
same as measured
13.22
1.37
4.71
1.37
same as measured
14.32
1.35
4.61
1.35
same as measured
0
5
10
15
Database-node CPU
(s per GiB of input)